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Report on
Where Do We Stand with Implicit Neural
Representations?
A Technical and Performance Survey

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Abstract

This report provides a critical examination of the comprehensive survey by Essakine et al. [1] on Implicit Neural Representations (INRs). The survey addresses a fundamental challenge in neural network research: how to effectively represent continuous signals using neural networks while overcoming the inherent spectral bias that causes standard architectures to favor low-frequency components. The authors introduce a systematic four-category taxonomy that organizes state-of-the-art INR methods based on their architectural innovations: activation function modifications, positional encoding enhancements, combined strategies, and network structure optimizations. Through extensive experiments spanning audio reconstruction, CT imaging, image denoising, super-resolution, and 3D occupancy reconstruction, the survey reveals important insights about method selection. No single approach dominates across all tasks; instead, methods combining spatial compactness, frequency compactness, and adaptivity—particularly INCODE, FINER, and FR—achieve consistently strong performance. A critical finding concerns computational efficiency: despite all methods sharing theoretical $O(1)$ query complexity, practical execution times vary by nearly six-fold, emphasizing that asymptotic analysis alone provides insufficient guidance for practitioners. This report synthesizes the survey’s contributions while critically examining its methodology, identifies limitations, and discusses implications for future research in neural signal representation.

Story Summary

1. What is the central question?

How do different Implicit Neural Representation (INR) methods compare in terms of performance across diverse tasks, and what architectural properties determine their effectiveness for representing continuous signals?

2. Why is this question important?

INRs have emerged as a promising paradigm for representing signals continuously rather than discretely, offering advantages like resolution independence and memory efficiency. However, the rapidly growing landscape of INR methods lacks systematic comparison, making it difficult for practitioners to select appropriate methods for specific applications.

3. What evidence/data (variables) are needed to answer this question?

Quantitative performance metrics (PSNR, SSIM, LPIPS, IoU, L2 loss) across multiple tasks (audio reconstruction, CT reconstruction, image denoising, super-resolution, 3D occupancy), computational timing data, and architectural property analysis (spatial compactness, frequency compactness, adaptivity).

4. What methods are used to get this evidence/data?

Systematic implementation of eight INR methods (SIREN, WIRE, GAUSS, FINER, Fourier Features, INCODE, MFN, FR) using default hyperparameters from original papers, trained and evaluated on standardized benchmarks using consistent hardware (RTX 4070 GPU).

5. What analyses must be applied to the data to answer the central question?

Comparative analysis of reconstruction quality metrics across methods and tasks, correlation analysis between architectural properties and performance outcomes, and computational efficiency profiling to assess practical deployment considerations.

6. What evidence/data (values for the variables) were obtained?

Comprehensive performance tables showing PSNR/SSIM/LPIPS values across all method-task combinations, timing measurements revealing $5.9\times$ variation in execution times (403s to 2370s for denoising), and property classifications for all eight methods.

7. What were the results of the analyses?

Methods combining spatial compactness, frequency compactness, and adaptivity (INCODE, FINER) achieved top-3 rankings across all five tasks, while FR excelled in four of five tasks. Methods lacking these properties never reached top-tier performance, and computational costs still varied dramatically despite identical theoretical complexity.

8. How did the analyses answer the central question?

The analyses revealed that no single method dominates all tasks, but architectural properties strongly predict performance. The combination of spatial compactness, frequency compactness, and adaptivity emerges as the key factor for consistent high performance across diverse applications.

9. What does this answer tell us about the broader field?

Future INR development should prioritize methods that integrate multiple representational properties rather than focusing on single innovations. Additionally, theoretical complexity analysis is insufficient for practical guidance; empirical evaluation remains essential for method selection.

10. Did the paper answer the question satisfactorily? Why (not)?

Yes, the paper provides comprehensive experimental evidence supporting its conclusions. However, limitations exist: only default hyperparameters were tested (potentially disadvantaging methods with more parameters), scalability to larger problems remains unexplored, and generalization to unseen data was not evaluated.

1 Introduction

Traditional approaches to signal representation rely on explicit discretization—images stored as pixel grids, 3D shapes encoded as voxel arrays, and audio signals captured as discrete samples. While effective for many applications, this paradigm faces fundamental limitations that become increasingly problematic as data complexity grows. Storage requirements scale directly with resolution, computational costs increase substantially with dimensionality, and working at arbitrary resolutions necessitates maintaining multiple versions of the same underlying data [5].

Implicit Neural Representations (INRs) offer a fundamentally different approach to this challenge. Rather than storing discrete samples, INRs encode data as continuous functions parameterized by neural networks [12]. When a value at any coordinate is required, the network is simply queried at that location. This paradigm provides several compelling advantages: resolution independence allows querying at arbitrary precision without retraining, memory efficiency scales with network parameters rather than output resolution, and full differentiability enables gradient-based optimization for complex inverse problems.

However, standard neural network architectures struggle to serve as effective implicit representations. Training a typical Multi-Layer Perceptron (MLP) with ReLU activations to represent a signal yields characteristically blurry, oversmoothed results. The network captures broad structural features but systematically fails to reproduce fine details. This phenomenon, termed “spectral bias” by Rahaman et al. [7], reflects a fundamental property of how neural networks learn: they naturally prioritize low-frequency components while struggling to capture high-frequency variations. Importantly, this is not merely a training issue that can be resolved with more iterations or better optimization—it stems from fundamental properties of how standard architectures interact with gradient descent dynamics.

The INR research community has responded to this challenge with diverse architectural innovations. Some approaches modify activation functions to incorporate periodic or localized behavior [12, 9, 8]. Others transform input coordinates before processing to enable high-frequency learning [13]. Hybrid strategies combine multiple techniques [10], while structural modifications alter fundamental network architecture [3, 2, 11]. Each approach addresses spectral bias differently, creating a rich but potentially confusing landscape with unclear relative merits.

The survey by Essakine et al. [1] provides the first systematic evaluation of these diverse approaches. By organizing methods into a clear taxonomy, implementing them consistently, and testing performance across five diverse tasks, the survey illuminates when and why different approaches excel. Rather than advocating for any single method, it reveals that method selection depends critically on task requirements and constraints. This report examines their findings in detail, assesses the methodology employed, and discusses implications for both practitioners and future research directions.

2 Background and Fundamental Concepts

2.1 Mathematical Formulation of INRs

Implicit Neural Representations formalize signal encoding as the problem of learning continuous functions from coordinates to values. Given an input coordinate $\mathbf{x} \in \mathbb{R}^{n_{in}}$, we seek to learn a mapping to outputs $a(\mathbf{x}) \in \mathbb{R}^{n_{out}}$ that satisfies some implicit equation involving the function and potentially its derivatives:

$$F(\mathbf{x}, a(\mathbf{x}), \nabla_{\mathbf{x}} a(\mathbf{x}), \nabla_{\mathbf{x}}^2 a(\mathbf{x}), \dots) = 0 \quad (1)$$

For image representation, coordinates $\mathbf{x} = (x, y)$ map to RGB pixel values. For 3D shape representation, coordinates $\mathbf{x} = (x, y, z)$ yield signed distance values indicating proximity to the

surface. The flexibility of this formulation enables INRs to represent diverse signal types within a unified framework.

The mapping is typically implemented using Multi-Layer Perceptrons (MLPs), where each layer computes:

$$\mathbf{y}_i = \sigma(\mathbf{W}_i \mathbf{y}_{i-1} + \mathbf{b}_i) \quad (2)$$

where σ denotes the activation function, $\mathbf{W}_i \in \mathbb{R}^{n_i \times n_{i-1}}$ represents the weight matrix, and $\mathbf{b}_i \in \mathbb{R}^{n_i}$ the bias vector for layer i . The input $\mathbf{y}_0$ corresponds to the coordinate $\mathbf{x}$, and the final layer produces the output without activation. The choice of activation function σ proves critical for determining what frequencies the network can effectively represent.

2.2 The Spectral Bias Problem

Standard MLPs with ReLU activations produce systematically poor results when used as implicit representations. Even deep, wide networks yield characteristically blurry outputs that capture only coarse features while missing fine details. Rahaman et al. [7] provided theoretical analysis demonstrating that neural networks preferentially learn low-frequency components—high-frequency details require exponentially more training iterations and often never fully emerge during practical training.

This spectral bias arises from properties of the Neural Tangent Kernel (NTK) that governs training dynamics in wide neural networks. Intuitively, ReLU networks create piecewise linear functions, and representing rapid oscillations requires many small linear pieces, making such functions inherently difficult to learn. The discontinuous derivative of ReLU at zero further complicates gradient flow for high-frequency patterns. Most subsequent INR research has focused specifically on overcoming this limitation through architectural modifications.

2.3 Key Properties of INRs

The survey identifies several important mathematical properties that distinguish INRs from traditional representations and influence their effectiveness:

Continuity and Differentiability: INRs model data as continuous, fully differentiable functions. This enables gradient-based optimization for inverse problems and provides smooth interpolation between sampled points without artifacts.

Resolution Independence: Output is not tied to any fixed resolution. The network can be queried at any coordinate without retraining, enabling applications like arbitrary-scale super-resolution from a single trained model.

Compactness: Memory requirements scale with the number of network parameters N rather than output resolution. This provides $O(N)$ storage regardless of how finely the function is queried.

Frequency Adaptivity: Through specialized encodings or adaptive activation functions, INRs can capture both low-frequency structure and high-frequency details, overcoming the inherent spectral bias of standard architectures.

Smoothness: Most INR architectures produce inherently smooth representations due to their activation functions, which proves advantageous for many natural signals but can limit sharp edge representation.

3 Taxonomy and Technical Analysis

The survey organizes INR methods into four distinct categories based on which architectural components they enhance. This taxonomy provides a useful framework for understanding the design space and comparing approaches.

3.1 Category (a): Activation Function Enhancement

The most direct approach to overcoming spectral bias involves replacing standard activation functions with alternatives designed to better capture high-frequency content.

SIREN [12]: This foundational method employs the sine function as activation:

$$\sigma(x) = \sin(\omega_0 x) \quad (3)$$

where $\omega_0 \approx 30$ serves as a frequency scaling hyperparameter. The periodic structure enables representation of high-frequency variations with smooth, well-behaved derivatives. SIREN requires careful weight initialization to maintain appropriate activation distributions across layers, but once trained produces high-quality reconstructions. The primary limitation is that the fixed frequency parameter may not suit all regions of a signal equally well.

GAUSS [8]: This approach employs a Gaussian activation function:

$$\sigma(x) = \exp(-(sx)^2) \quad (4)$$

where s controls the width. Gaussians provide both spatial and frequency localization, making them well-suited for smooth, localized features. However, the lack of periodicity limits effectiveness for representing sharp edges and high-frequency oscillations.

WIRE [9]: Building on insights from both periodic and Gaussian approaches, WIRE uses Gabor wavelets:

$$\sigma(x) = \exp(-(sx)^2 + i\omega x) \quad (5)$$

This complex-valued activation combines Gaussian envelope localization with sinusoidal frequency content, achieving optimal joint time-frequency localization according to uncertainty principle bounds. The approach excels at representing signals with spatially-varying frequency content but introduces additional computational overhead and two hyperparameters requiring careful tuning.

FINER [4]: This recent method addresses SIREN's limitation of fixed frequency by making the effective frequency input-dependent:

$$\sigma(x) = \sin(\omega_0(|x| + 1) \cdot x) \quad (6)$$

The effective frequency now depends on the input magnitude, allowing different neurons to specialize in different frequency ranges based on their bias initialization. This provides adaptivity that fixed-frequency methods lack, enabling representation of broader frequency spectra without manual tuning.

3.2 Category (b): Positional Encoding

Rather than modifying activations, positional encoding approaches transform input coordinates before processing to enable high-frequency learning even with standard activations.

Fourier Features [13]: This approach maps inputs through a Fourier feature transformation:

$$\gamma(\mathbf{x}) = [\cos(2\pi\mathbf{B}\mathbf{x}), \sin(2\pi\mathbf{B}\mathbf{x})]^T \quad (7)$$

where $\mathbf{B} \sim \mathcal{N}(0, \sigma^2)$ is a random Gaussian matrix. This transformation lifts inputs into a higher-dimensional space where high-frequency patterns become learnable even by ReLU networks. The primary limitation is that features are fixed after initialization, preventing adaptation during training.

3.3 Category (c): Combined Strategies

Some methods integrate both activation function and positional encoding enhancements for potentially synergistic benefits.

TRIDENT [10]: This approach uses Fourier feature mapping in the first layer followed by Gaussian activations in subsequent layers. The combination enables both high-order feature representation and broad frequency spectrum coverage while maintaining spatial localization through Gaussian windowing. The approach introduces additional hyperparameter complexity but provides a unified framework.

3.4 Category (d): Network Structure Optimization

Rather than modifying components, these methods alter fundamental network architecture.

INCONE [3]: This method uses dynamically modulated activations:

$$\sigma(x) = a \sin(b\omega_0 x + c) + d \quad (8)$$

where parameters a, b, c, d are predicted by a separate “harmoniser” network during training. This enables real-time adaptation of activation characteristics to match signal properties, providing unprecedented flexibility. The computational overhead of the harmoniser network is significant but justified by consistent performance gains.

MFN (Multiplicative Filter Networks) [2]: This architecture replaces additive layer compositions with Hadamard (element-wise) products, creating fundamentally different learning dynamics. MFN supports both sinusoidal and Gabor wavelet activation functions within this multiplicative framework. While capable of representing complex patterns, training requires careful hyperparameter selection.

FR (Fourier Reparameterized Training) [11]: This approach reparameterizes weight matrices using Fourier bases:

$$\mathbf{y}_i = \sigma(\mathbf{\Lambda}_i \mathbf{B}_i \mathbf{y}_{i-1} + \mathbf{b}_i) \quad (9)$$

where $\mathbf{\Lambda}_i$ is learnable and $\mathbf{B}_i$ contains Fourier basis functions. This directly incorporates frequency information into the architecture, explicitly addressing spectral bias at the weight level.

3.5 Properties Comparison

The survey identifies three key architectural properties that correlate with performance:

- **Spatial Compactness:** The ability to represent localized spatial features without global interference.
- **Frequency Compactness:** The ability to isolate specific frequency bands without spectral leakage.
- **Adaptivity:** The ability to adjust representational characteristics during training based on signal properties.

Table 1 summarizes how methods compare across these properties.

Methods combining all three properties (INCONE, FINER) tend to perform consistently well across diverse tasks, while methods lacking multiple properties show more variable performance.

Table 1: Comparison of INR methods across key architectural properties.

Method	Category	Spatial	Frequency	Adaptivity	Hyperparams
SIREN	(a)	✓	×	×	1
GAUSS	(a)	✓	✓	×	1
WIRE	(a)	✓	✓	×	2
FINER	(a)	✓	✓	✓	2
Fourier Features	(b)	×	×	×	1
TRIDENT	(c)	✓	✓	×	2
INCODE	(d)	✓	✓	✓	1
MFN	(d)	×	×	×	2
FR	(d)	×	×	✓	1

4 Experimental Results

The survey evaluates methods across five diverse tasks using publicly available implementations with default hyperparameters on consistent hardware (RTX 4070 GPU with 8GB RAM). This standardized evaluation enables fair comparison while acknowledging that hyperparameter tuning could potentially improve individual method performance.

4.1 Audio Reconstruction

The audio reconstruction task involves learning a continuous representation of a 6-second Bach musical piece. This 1D task tests fundamental signal representation capability with relatively simple spatial structure but complex frequency content. All methods use a 5-layer network trained for 2000 iterations with a first-layer scaling factor of 30.

Table 2: Audio reconstruction performance (L2 loss, lower is better).

Method	L2 Loss
FR	0.00000415
INCODE	0.00000736
FINER	0.0000121
SIREN	0.0000394
Fourier Features	0.0141
GAUSS	0.0183
MFN	0.0241
WIRE	0.0986

FR and INCODE substantially outperform other methods, achieving losses approximately four orders of magnitude lower than weaker approaches. FR’s explicit Fourier basis reparameterization directly addresses the frequency representation challenge, while INCODE’s dynamic modulation adapts to the signal’s varying frequency content. Notably, WIRE performs poorly despite its Gabor wavelet design, suggesting that dual localization properties may not transfer directly to 1D audio signals.

4.2 CT Image Reconstruction

CT reconstruction evaluates medical imaging applications where images must be recovered from limited projection measurements. The survey tests reconstruction quality across varying numbers

of projections (20, 50, 100, 150, 200, 300), using a 2-layer network with 300 hidden features trained for 5000 iterations.

Table 3: CT reconstruction at 20 and 300 projections.

Method	20 Projections		300 Projections	
	PSNR	SSIM	PSNR	SSIM
FR	25.58	0.738	30.40	0.910
INCODE	24.38	0.651	34.76	0.953
FINER	25.53	0.731	32.24	0.927
SIREN	20.85	0.421	32.71	0.928
Fourier Features	25.64	0.780	26.94	0.794
GAUSS	22.21	0.548	27.90	0.848
MFN	21.10	0.402	33.63	0.935
WIRE	20.73	0.417	30.95	0.902

Results reveal an important data-regime dependency. FR excels with sparse measurements (20-50 projections), achieving highest quality when data is limited. However, INCODE dominates with abundant measurements (150+ projections), reaching 34.76 dB PSNR at 300 projections. This suggests different methods suit different clinical scenarios: FR for rapid low-dose imaging, INCODE for high-quality diagnostic reconstruction.

4.3 Image Denoising

Image denoising tests the ability to recover clean images from observations corrupted by Poisson and Gaussian noise, using a 2-layer network with 256 hidden features trained for 2000 iterations.

Table 4: Image denoising performance and computational time.

Method	PSNR (dB) $\uparrow$	Time (s) $\downarrow$
INCODE	29.63	2370
FR	29.47	752
FINER	29.05	643
SIREN	28.60	482
WIRE	28.74	1485
GAUSS	28.09	709
MFN	28.22	1280
Fourier Features	27.16	403

This task reveals critical quality-efficiency trade-offs. INCODE achieves the highest PSNR (29.63 dB) but requires nearly 6x longer than the fastest method. FR and FINER offer compelling alternatives with near-peak quality (within 0.6 dB of INCODE) at approximately 3-4x faster execution. For time-critical applications, SIREN provides reasonable quality (28.60 dB) at moderate computational cost.

4.4 Single Image Super-Resolution

Super-resolution tests the ability to reconstruct high-resolution images from low-resolution inputs across four scale factors (2x, 4x, 8x, 16x). This task particularly stresses frequency representation capability at higher upscaling factors.

Table 5: Super-resolution at 2x and 16x upscaling factors.

Method	2x Resolution			16x Resolution		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
INCORE	29.56	0.896	0.176	22.91	0.638	0.715
FINER	29.50	0.892	0.191	23.39	0.667	0.665
FR	29.10	0.879	0.243	23.40	0.682	0.637
SIREN	29.00	0.877	0.241	19.09	0.509	0.780
MFN	29.28	0.890	0.203	16.61	0.290	0.934
GAUSS	28.08	0.851	0.324	17.14	0.349	0.868
WIRE	28.91	0.874	0.252	18.06	0.422	0.773
Fourier Features	26.31	0.767	0.428	20.65	0.518	0.702

Performance differences become dramatic at extreme upscaling factors. At 2x, most methods perform comparably (within 3 dB PSNR). At 16x, methods lacking frequency compactness (GAUSS, MFN) collapse catastrophically, with MFN dropping to 16.61 dB—a degradation of over 12 dB from its 2x performance. FR and FINER maintain relative stability, demonstrating that adaptivity becomes critical for extreme reconstruction tasks.

4.5 3D Occupancy Reconstruction

The 3D occupancy task tests geometric representation capability using a complex dragon mesh with over 2.7 million points and 5.5 million triangles. Methods must learn to classify points as inside or outside the shape.

Table 6: 3D occupancy reconstruction (IoU, higher is better).

Method	IoU
FINER	0.99628
INCORE	0.99564
SIREN	0.99552
GAUSS	0.99510
WIRE	0.99454
Fourier Features	0.99424
FR	0.99136
MFN	0.97540

Most methods achieve high IoU values (above 0.99), but the three worst performers—MFN, FR, and Fourier Features—share a notable property: they all lack frequency compactness. This suggests that frequency compactness is particularly important for representing fine geometric details in 3D reconstruction tasks.

4.6 Cross-Task Performance Summary

Table 7 synthesizes performance rankings across all tasks.

INCORE and FINER achieve top-3 performance across all five tasks, demonstrating the value of combining spatial compactness, frequency compactness, and adaptivity. FR excels in four of five tasks, while SIREN—the foundational INR method—maintains competitive performance in three tasks despite being the simplest approach. Methods lacking multiple key properties never achieve top-tier performance.

Table 7: Top-3 rankings across all evaluated tasks.

Method	Audio	CT	Denoise	SR	3D	Total
INCODE	✓	✓	✓	✓	✓	5/5
FINER	✓	✓	✓	✓	✓	5/5
FR	✓	✓	✓	✓	×	4/5
SIREN	✓	×	✓	×	✓	3/5
Others	×	×	×	×	×	0/5

5 Critical Assessment

5.1 Methodological Strengths

The survey provides several important contributions to the INR field:

Systematic Taxonomy: The four-category organization (activation functions, positional encoding, combined strategies, network structure) provides a clear conceptual framework that helps researchers understand the design space and relationships between approaches. This taxonomy will likely influence how future work is positioned and compared.

Fair Experimental Comparison: Using consistent hardware, default hyperparameters, and identical training protocols enables unbiased comparison that is often lacking in individual method papers. Each method is evaluated on its own terms as published, rather than with potentially unfair tuning.

Task Diversity: Testing across five fundamentally different applications (1D audio, 2D imaging, 3D geometry) reveals performance patterns that narrow single-task evaluations would miss. The observation that different methods excel in different regimes provides practical guidance for practitioners.

Computational Realism: Including timing measurements alongside quality metrics addresses practical concerns that purely quality-focused evaluations often ignore. The finding that execution times vary 5.9x despite identical theoretical complexity is particularly valuable.

Property Analysis: Identifying spatial compactness, frequency compactness, and adaptivity as key determinants of performance provides actionable insights for both method selection and future architecture design.

5.2 Limitations and Gaps

Despite its contributions, the survey has notable limitations:

Hyperparameter Sensitivity: Using only default hyperparameters potentially disadvantages methods with more parameters (WIRE, FINER, MFN, TRIDENT with 2 hyperparameters each) compared to single-parameter methods. Methods with more tuning options might perform better with optimization.

Scalability Analysis: All experiments use moderate-scale problems. Behavior on gigapixel images, massive 3D models, or long audio sequences remains unexplored. Scalability is critical for many real-world applications.

Generalization Evaluation: The survey focuses exclusively on single-signal fitting, where each network encodes one specific signal. How methods generalize to encode multiple signals or handle unseen data—increasingly important for practical applications—is not evaluated.

Training Dynamics: Convergence behavior, optimization stability, and initialization sensitivity are not systematically analyzed. Understanding why certain methods train more reliably would provide valuable guidance.

Theoretical Foundations: Limited formal analysis explains why certain architectures improve frequency handling. Principled design frameworks connecting architecture to representational capacity would strengthen both understanding and future development.

Robustness Analysis: How methods handle distribution shift, adversarial inputs, or unexpected data patterns remains unexplored. Robustness is critical for deployment in real-world settings where data may not match training assumptions.

5.3 Missing Context

The survey focuses on architectural innovations within the core INR paradigm but does not extensively cover connections to related areas. Neural Radiance Fields (NeRF) [6], which build upon INR foundations for novel view synthesis, represent a major application area with potentially different requirements. Coordinate-based graphics networks for animation and generative applications also present distinct challenges. Understanding what lessons transfer across these domains would provide valuable perspective for researchers working at application boundaries.

Recent work on transformers for INRs and diffusion models for continuous representations receives only brief mention. These emerging directions may significantly reshape the field, and understanding their relationship to traditional INR methods would be valuable for researchers choosing directions for future work.

6 Implications and Recommendations

6.1 Practical Guidance for Method Selection

Based on the survey’s findings, the following recommendations emerge for practitioners:

General-Purpose Applications: SIREN provides a reliable baseline with minimal tuning requirements. Its single hyperparameter and well-understood behavior make it a safe starting point for most applications.

Quality-Critical Applications: When computational resources permit, INCODE consistently achieves the highest reconstruction quality. The additional complexity of its harmoniser network is justified for applications where quality is paramount.

Quality-Efficiency Trade-offs: FR and FINER offer compelling alternatives when both quality and speed matter. They achieve near-peak quality at significantly reduced computational cost, making them suitable for interactive applications or large-scale processing.

Task-Specific Recommendations:

- **Audio:** FR or INCODE for highest fidelity
- **Sparse Medical Imaging:** FR for limited-projection scenarios
- **Dense Medical Imaging:** INCODE for high-quality diagnostic reconstruction
- **Extreme Super-Resolution:** FR or FINER for robustness at high upscaling factors
- **3D Geometry:** FINER or INCODE for detailed surface reconstruction

6.2 Directions for Future Research

The survey’s findings suggest several promising research directions:

Unified Frameworks: Developing methods that systematically combine activation function, positional encoding, and structural innovations while reducing hyperparameter complexity could yield both simpler and more effective architectures.

Automated Architecture Search: Given the strong relationship between architectural properties and performance, neural architecture search techniques tailored for INRs could automatically discover optimal configurations for specific tasks.

Computational Efficiency: Bridging the gap between theoretical $O(1)$ complexity and practical 5.9x execution time variation through distillation, pruning, or specialized hardware implementations would expand the applicability of high-quality methods.

Generalization Studies: Understanding how INRs trained on multiple examples generalize to unseen data is increasingly important as the field moves beyond single-signal fitting toward more practical meta-learning applications.

Theoretical Foundations: Developing formal frameworks that connect architectural choices to representational capacity, learning dynamics, and convergence properties would provide principled guidance for future design.

7 Conclusion

The survey by [1] offers a comprehensive and systematic examination of implicit neural representation methods and shows that there is no universally best architecture; method selection must reflect task requirements and constraints. Its taxonomy and unified benchmark make the trade-offs between reconstruction quality and efficiency explicit.

The central insight is that combining spatial compactness, frequency compactness, and adaptivity consistently drives strong performance: INCODE and FINER reach top-3 rankings across all five tasks, and FR excels in four of them by explicitly encoding frequency information, while methods lacking these properties rarely achieve top-tier results. The observation that nominally $O(1)$ methods still differ by up to $5.9\times$ in runtime underlines the need for empirical evaluation on realistic hardware, not just asymptotic analysis. Overall, the survey sets a methodological standard for INR architecture evaluation and is likely to remain a key reference as continuous neural representations are adopted more widely in vision, graphics, medical imaging, and scientific computing.

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